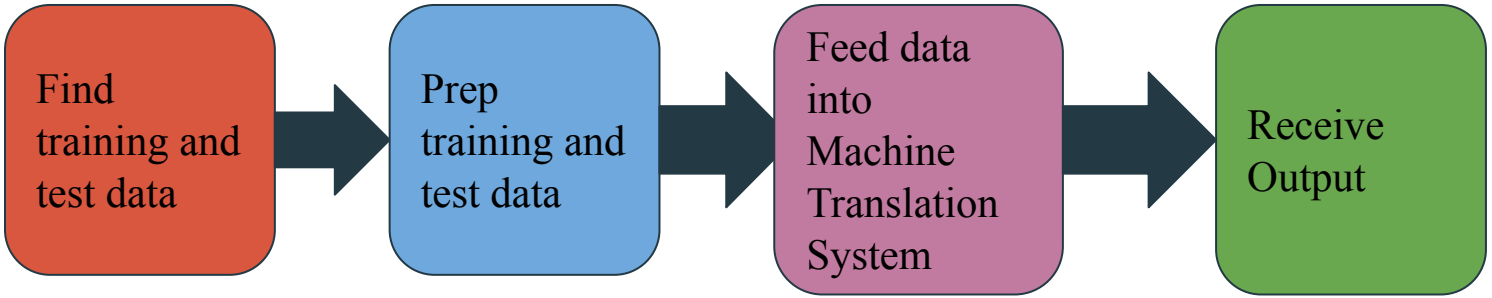
Senior Design Powerpoint Presentation

Anurag Neti

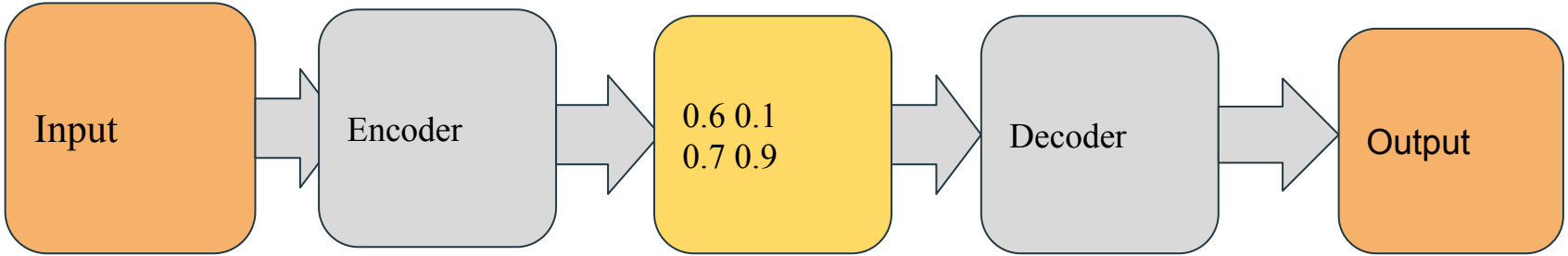
Project Overview

1. For my Senior Design, I decided to implement a Neural Machine Translation to translate between English and Japanese.
2. The technology leveraged Neural Networks and Deep Learning to perform this task.

Project Pipeline



Machine Translation Pipeline



Challenges faced

1. Research
2. Finding training data
3. Preparing the training data

Research

Primary Sources of research include

1. YouTube (tutorials)
2. Medium (tutorials)

Finding Training Data

1. The machine translation system required parallel corpora.

   Parallel Corpora - Text placed alongside its translation in another language.

2. A shell script was executed to download this parallel corpora once it was located.

Preparing Training Data

1. To ensure the classifier i.e machine translation system was actually learning from the training data, and recognizing patterns in both Japanese and English, the sentences within the corpora needed to be randomized while maintaining the same relative order.
2. I implemented a shell script to achieve this task.

The Neural Machine Translation System

1. The translation system was designed using Marian NMT, a C++ based Neural Machine Translation framework.
2. Once the training was located was found and normalized, Marian NMT learnt from it and produced the Japanese translation of an English text.